# NORMALIZATION

# Definition

- It is a formal process for deciding which attributes should be grouped together in a relation.
- It is a process of successively reducing relations with anomalies to produce smaller; well-structured relations.

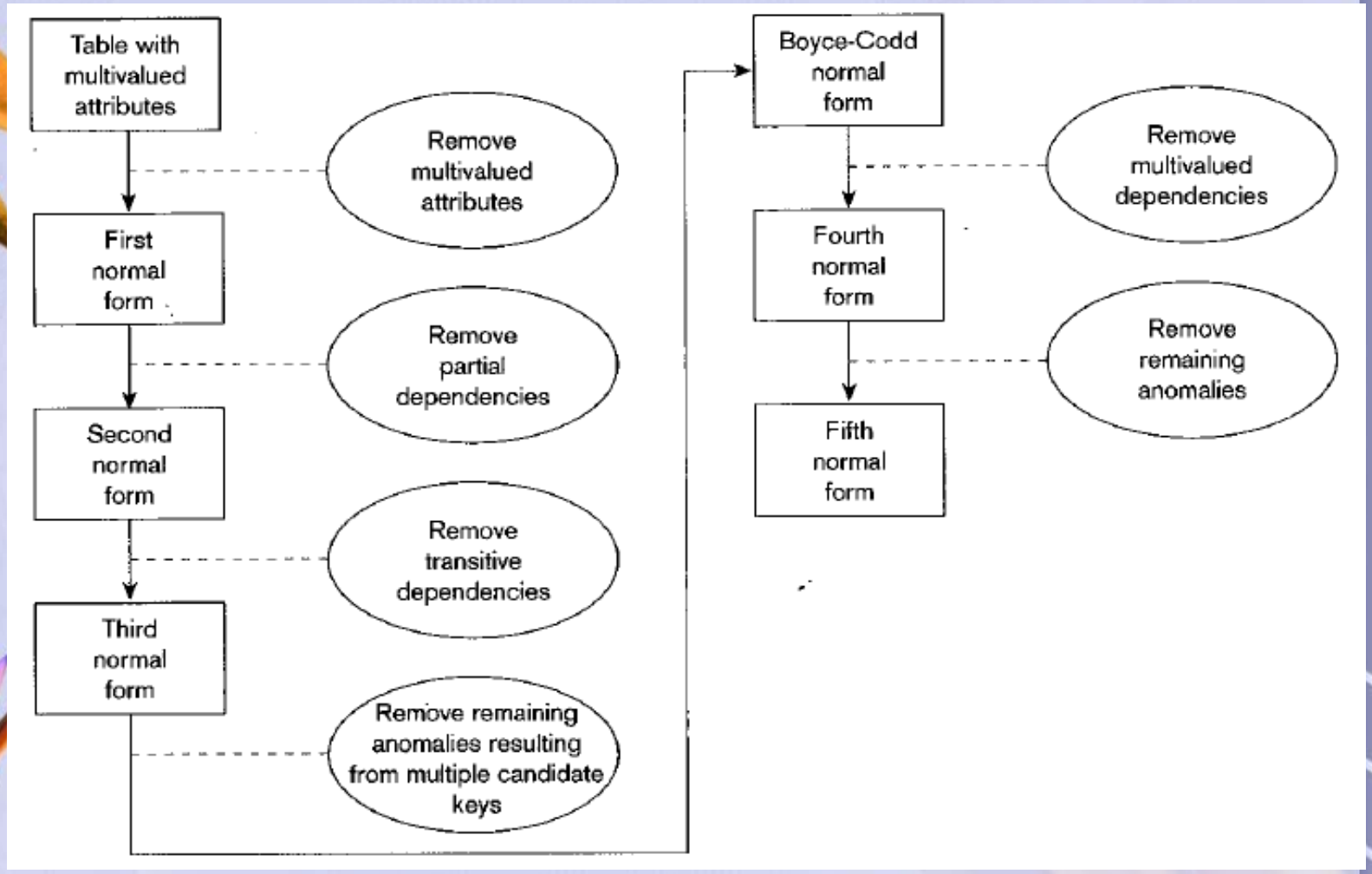
# Goals of Normalization

- Minimize data redundancy, thereby avoiding anomalies and conserving storage space
- Simplify the enforcement of referential integrity constraints.
- Make it easier to maintain data (insert, update, and delete)
- Provide a better design that is an improved representation of the real world.

# Steps in Normalization

- Normalization can be accomplished and understood in stages, each of which corresponds to a normal form.
- Normal form is a state of a relation that results from applying simple rules regarding functional dependencies (or relationships between attributes) to that relation.

# Steps in Normalization



# First Normal Form (1NF)

- Any multi-valued attributes (also called repeating groups) have been removed, so there is a single value at the intersection of each row and column of the table.

# First Normal Form (1NF)

(a) Table with repeating groups

| Emp_ID | Name | Dept_Name | Salary | Course_Title | Date_Completed |
|---|---|---|---|---|---|
| 100 | Margaret Simpson | Marketing | 48,000 | SPSS | 6/19/200X |
| | | | | Surveys | 10/7/200X |
| 140 | Alan Beeton | Accounting | 52,000 | Tax Acc | 12/8/200X |
| 110 | Chris Lucero | Info Systems | 43,000 | Visual Basic | 1/12/200X |
| | | | | C++ | 4/22/200X |
| 190 | Lorenzo Davis | Finance | 55,000 | | |
| 150 | Susan Martin | Marketing | 42,000 | SPSS | 6/16/200X |
| | | | | Java | 8/12/200X |

(b) EMPLOYEE2 relation

EMPLOYEE2

| Emp_ID | Name | Dept_Name | Salary | Course_Title | Date_Completed |
|---|---|---|---|---|---|
| 100 | Margaret Simpson | Marketing | 48,000 | SPSS | 6/19/200X |
| 100 | Margaret Simpson | Marketing | 48,000 | Surveys | 10/7/200X |
| 140 | Alan Beeton | Accounting | 52,000 | Tax Acc | 12/8/200X |
| 110 | Chris Lucero | Info Systems | 43,000 | Visual Basic | 1/12/200X |
| 110 | Chris Lucero | Info Systems | 43,000 | C++ | 4/22/200X |
| 190 | Lorenzo Davis | Finance | 55,000 | | |
| 150 | Susan Martin | Marketing | 42,000 | SPSS | 6/19/200X |
| 150 | Susan Martin | Marketing | 42,000 | Java | 8/12/200X |

# First Normal Form (1NF)

INVOICE data (Pine Valley Furniture Company)

| Order_ID | Order_ Date | Customer_ ID | Customer_ Name | Customer_ Address | Product_ID | Product_ Description | Product_ Finish | Unit_ Price | Ordered_ Quantity |
|---|---|---|---|---|---|---|---|---|---|
| 1006 | 10/24/2006 | 2 | Value Furniture | Plano, TX | 7 | Dining Table | Natural Ash | 800.00 | 2 |
| | | | | | 5 | Writer's Desk | Cherry | 325.00 | 2 |
| | | | | | 4 | Entertainment Center | Natural Maple | 650.00 | 1 |
| 1007 | 10/25/2006 | 6 | Furniture Gallery | Boulder, CO | 11 | 4-Dr Dresser | Oak | 500.00 | 4 |
| | | | | | 4 | Entertainment Center | Natural Maple | 650.00 | 3 |

# First Normal Form (1NF)

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|---|---|---|---|---|---|---|---|---|---|
| 1006 | 10/24/2006 | 2 | Value Furniture | Plano, TX | 7 | Dining Table | Natural Ash | 800.00 | 2 |
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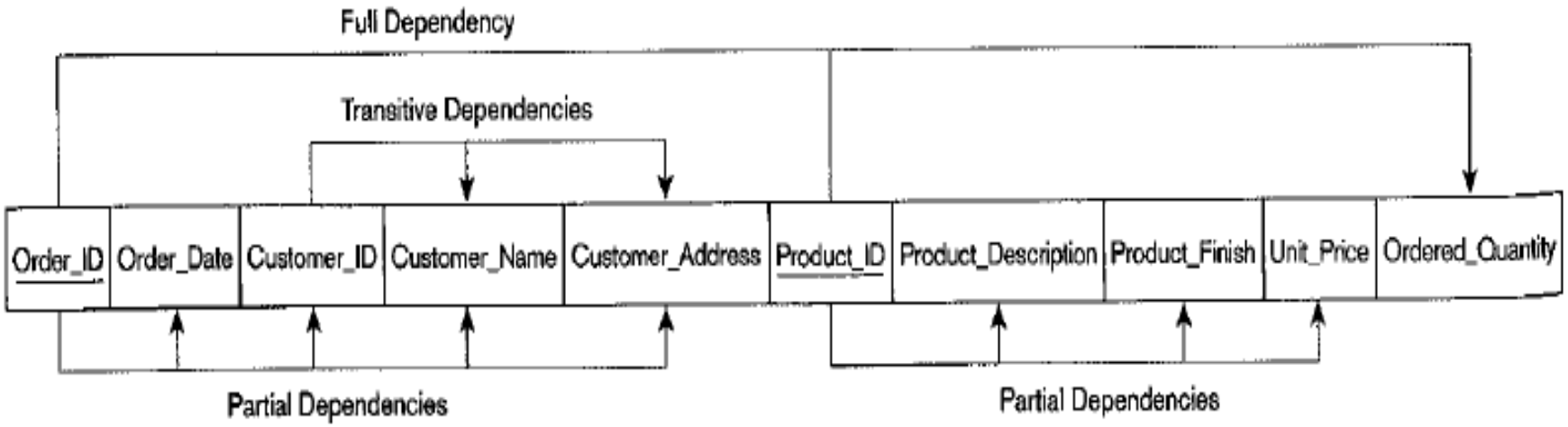
| 1007 | 10/25/2006 | 6 | Furniture Gallery | Boulder, CO | 11 | 4-Dr Dresser | Oak | 500.00 | 4 |
| 1007 | 10/25/2006 | 6 | Furniture Gallery | Boulder, CO | 4 | Entertainment Center | Natural Maple | 650.00 | 3 |

# Second Normal Form (2NF)

- Any partial dependencies have been removed
  - Partial functional dependency exists when a non-key attribute is functionally dependent on part (but not all) of the primary key.

Functional dependency diagram for INVOICE



# Second Normal Form (2NF)

- To convert a relation with partial dependencies to second normal form, the following steps are required:
  - Create a new relation for each primary key attribute that is a determinant in a partial dependency. The attribute is the primary key in the new relation.
  - Move the non-key attributes that are dependent on this primary key attribute (or attributes) from the old relation to the new relation.

# Second Normal Form (2NF)

| Order_ID | Order_ Date | Customer_ ID | Customer_ Name | Customer_ Address | Product_ID | Product_ Description | Product_ Finish | Unit_ Price | Ordered_ Quantity |
|---|---|---|---|---|---|---|---|---|---|
| 1006 | 10/24/2006 | 2 | Value Furniture | Plano, TX | 7 | Dining Table | Natural Ash | 800.00 | 2 |
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| 1007 | 10/25/2006 | 6 | Furniture Gallery | Boulder, CO | 4 | Entertainment Center | Natural Maple | 650.00 | 3 |

- partial dependencies

  Order_ID → Order_Date, Customer_ID, Customer_Name, Customer_Address

  Product_ID → Product_Description, Product_Finish, Unit_Price

# Second Normal Form (2NF)

| Order_ID | Product_ID | Ordered_Quantity |
|---|---|---|

ORDER_LINE

| Product_ID | Product_Description | Product_Finish | Unit_Price |
|---|---|---|---|

PRODUCT

| Order_ID | Order_Date | Customer_ID | Customer_Name | Customer_Address |
|---|---|---|---|---|

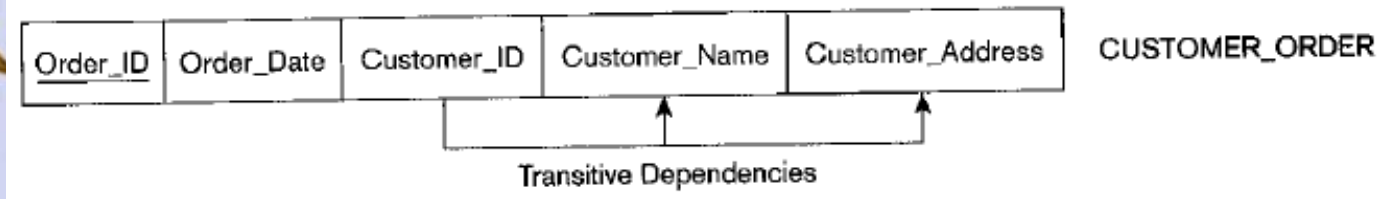
CUSTOMER_ORDER

Transitive Dependencies

# Third Normal Form (3NF)

- Any transitive dependencies have been removed.
  - A transitive dependency in a relation is a functional dependency between two (or more) nonkey attributes.

# Third Normal Form (3NF)

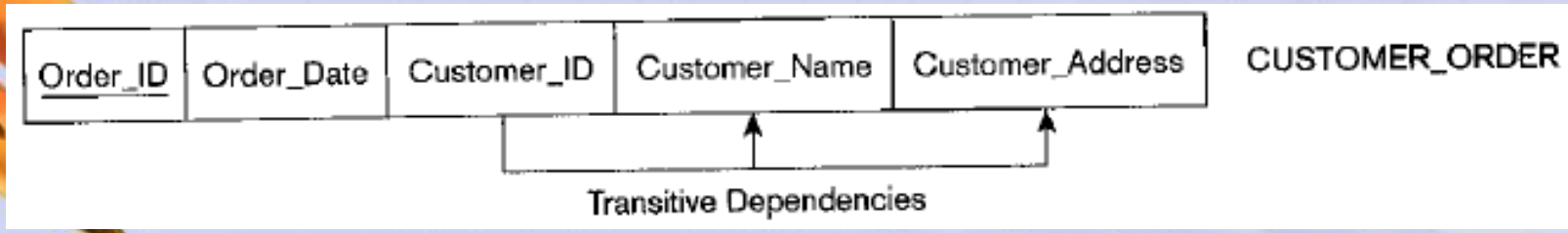


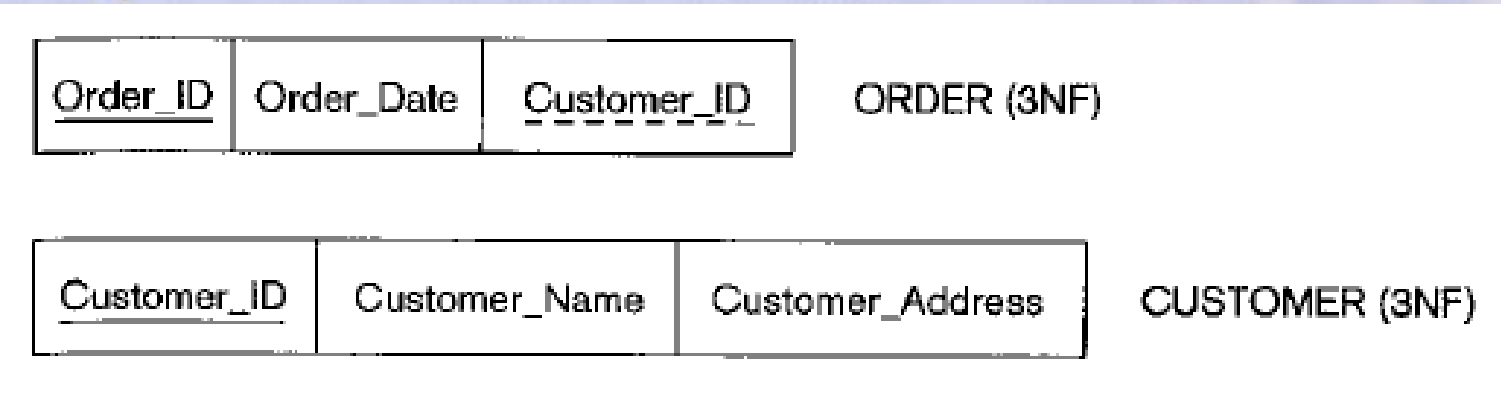

- transitive dependencies :
  - Customer_ID → Customer_Name
  - Customer_ID → Customer_Address

# Third Normal Form (3NF)

- Removing Transitive Dependencies
  - For each nonkey attribute (or set of attributes) that is a determinant in a relation, create a new relation. That attribute (or set of attributes) becomes the primary key of the new relation.
  - Move all of the attributes that are functionally dependent on the attribute from the old to the new relation.
  - Leave the attribute (which serves as a primary key in the new relation) in the old relation to serve as a foreign key that allows you to associate the two relations.

# Third Normal Form (3NF)



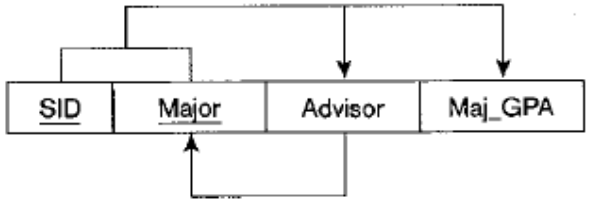
# Boyce-Codd Normal Form (3NF)

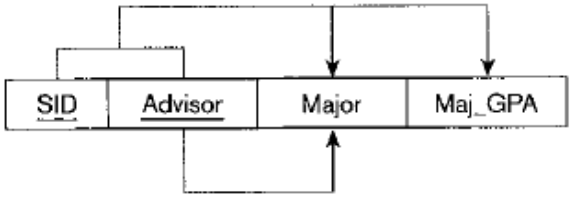
- Any remaining anomalies that result from functional dependencies have been removed

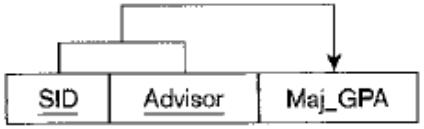
# Boyce-Codd Normal Form (3NF)

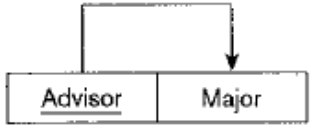
STUDENT_ADVISOR

| SID | Major | Advisor | Maj_GPA |
|---|---|---|---|
| 123 | Physics | Hawking | 4.0 |
| 123 | Music | Mahler | 3.3 |
| 456 | Literature | Michener | 3.2 |
| 789 | Music | Bach | 3.7 |
| 678 | Physics | Hawking | 3.5 |




STUDENT

| SID | Advisor | Maj_GPA |
|---|---|---|
| 123 | Hawking | 4.0 |
| 123 | Mahler | 3.3 |
| 456 | Michener | 3.2 |
| 789 | Bach | 3.7 |
| 678 | Hawking | 3.5 |

ADVISOR

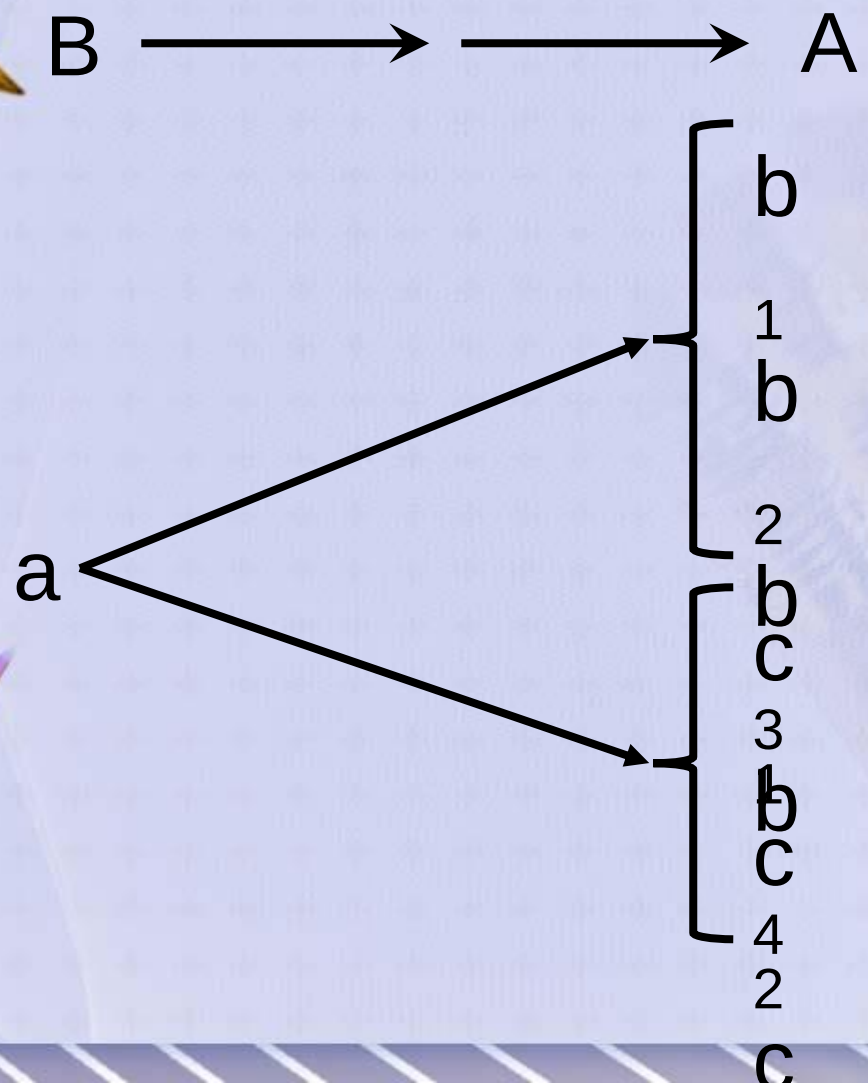
| Advisor | Major |
|---|---|
| Hawking | Physics |
| Mahler | Music |
| Michener | Literature |
| Bach | Music |

# Fourth Normal Form

No multivalued dependencies

- A multivalued dependency of column B on column A occurs when a table has a key with three or more attributes, (A, B, C) and
  - each value of A is associated with a collection of values of B
  - this collection of values is independent of C

# B is multidependent on A



# Example: multivalued dependence

*Faculty have advisee assignments for one major and committee assignments*

[FacID | AdviseeID | FacultyComm]

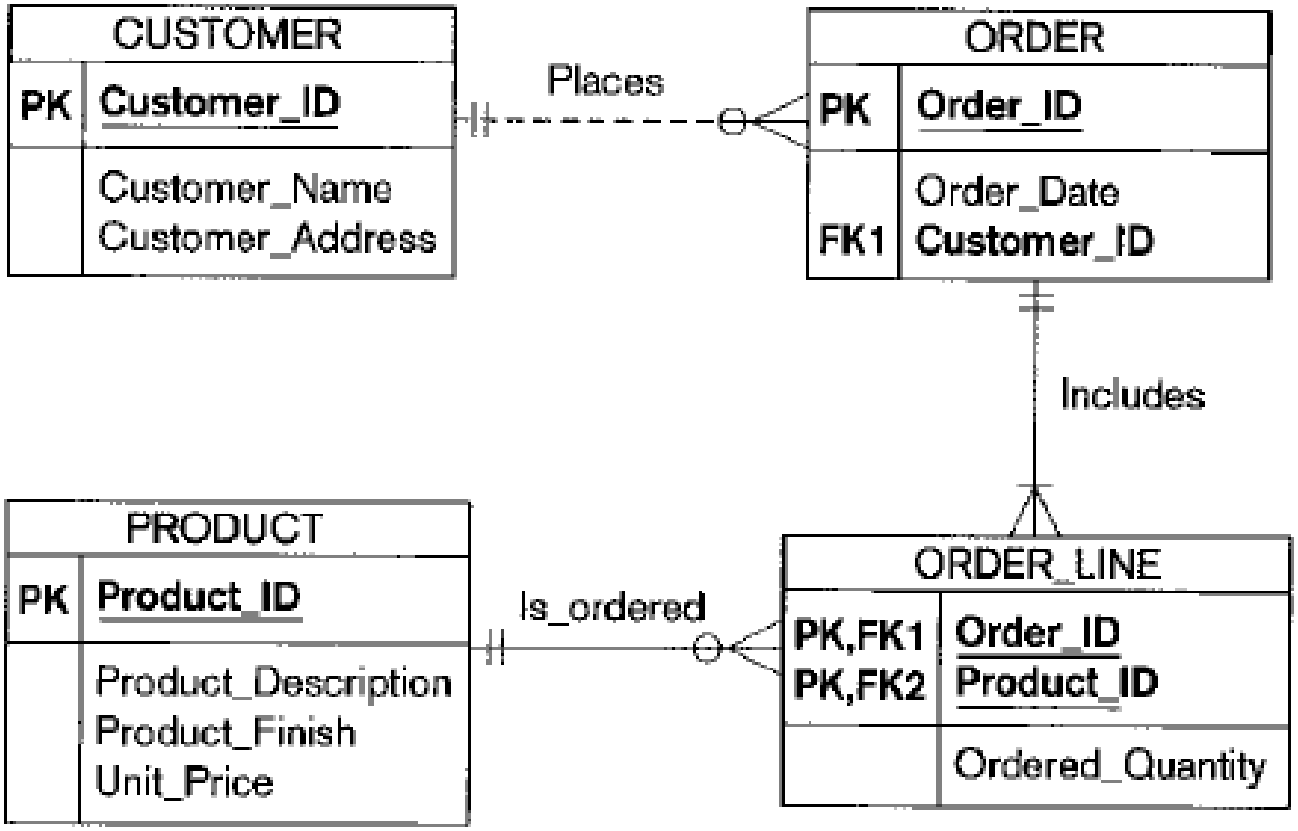
[FacultyID]

[FacID | AdviseeID]  [FacID | FacultyComm]

# Fifth Normal Form

Any remaining anomalies have been removed.

# Relational Schema



# Normalization

**1NF** Keys & no repeating groups

**2NF** 1NF & all attributes depend on all key components

**3NF** 2NF & all determinants are candidate keys

**4NF** 3NF & no multivalued dependencies